1. An object is 24 cm in front of a converging lens with a focal length of 12 cm. Use ray tracing to determine the location of the image. Is the image upright or inverted?

2. An object is 6 cm in front of a converging lens with a focal length of 10 cm. Use ray tracing to determine the location of the image. Is the image upright or inverted? What is the magnification?

3. An object is 15 cm in front of a diverging lens with a focal length of -15 cm. Use ray tracing to determine the location of the image. Is the image upright or inverted? Is the image real or virtual?

4. A double convex lens is constructed from glass with n=1.5. The left surface has a radius of curvature of 40 cm. The right surface has a radius of curvature of 24 cm. What is the focal length of the lens?

5. A double concave lens is constructed from glass with n=1.5. The two surfaces each have a radius of curvature of 40 cm. What is the focal length of the lens?

6. A 2.0-cm-tall object is 40 cm away from a converging lens that has a 20 cm focal length.

A. Use ray tracing to find the position and height of the image. To do this accurately, use a ruler or paper with a grid. Determine the image distance and image height by making measurements on your diagram

B. Calculate the image position, magnification, and height. Compare with your answer on part **A.**

7. In recent years, physicists have learned to create *metamaterials* – engineered materials not found in nature – with negative indicies of refraction. It's not yet possible to form a lens from a material with a negative index of refraction, but researchers are optimistic! Suppose you had a planoconvex lens (flat on one side, a 15 cm radius of curvature on the other) that is made from a metamaterial with n=-1.25. If you place an object 12 cm from the this lens:

A. What type of image will be formed?

B. Where will the image be located?

8. Play around with the lens sandbox in the Day 1.6 notes to gain a better understanding of ray tracing. Then use GeoGebra to answer: for a focal length of f =-80 cm, for which object distance is the image located where the actual object is? Is this image identical to the object?

GeoGebra hints (`https://www.geogebra.org/graphing`)

- Type the focal length: `f = -80`
- Type the image distance as a function of the object distance $x = d_o$ (from the thin-lens equation):
  `img(x) = 1 / (1/f - 1/x)` (don't name it `i`: GeoGebra reserves i)
- "The image is where the object is" means $d_i = -d_o$ (same distance, on the object side). Type `g(x) = -x`
- Right-click the graph → *Graphics* and set the window to $0 \leqslant x \leqslant 300$, $-300 \leqslant y \leqslant 300$. Then use `Intersect(img, g)` or the **Intersect** tool.
- If GeoGebra finds no intersection, zoom in toward the origin. What happens to the two graphs as $x \to 0$?
- For the last question, type `M(x) = -img(x)/x`. What does M approach?

9. Go to `https://phet.colorado.edu/en/simulations/geometric-optics`. What happens to the image location as (1) the index of refraction and (2) the radius of curvature is varied?